Spatial Data Quality Assurance Multiple Choice (Assignment for Module 3, partly also for Module 2)

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You are kindly asked to answer the questions and send the document to the e-mail address: Nello.RizzoNaudi@pa.org.mt by , 8 April 2026, so we can provide you with feedback on time.

1. **Which of the following is NOT a component of spatial data quality?**
   - ☐ Positional Accuracy
   - ☐ Attribute Accuracy
   - ☒ Colour Coding
   - ☐ Logical Consistency

2. **What is your opinion regarding the statement: "*Spatial data is considered accurate if it looks correct on a map*"?**
   - ☐ True
   - ☒ False

3. **What does 'fit for purpose' mean in the context of spatial data?**
   - ☐ The data is beautiful and detailed
   - ☒ The data meets the needs of its intended use
   - ☐ The data includes all countries
   - ☐ The data was collected recently

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4. **Which error type occurs when attributes don't match the spatial features?**
   - ☐ Topological error
   - ☒ Attribute error
   - ☐ Positional error
   - ☐ Metadata error

5. **What is your opinion regarding the statement: "*Topology rules help maintain spatial relationships between features, such as adjacency and connectivity*"?**
   - ☒ True
   - ☐ False

6. **What is the primary purpose of a quality assurance (QA) process in spatial data management?**
   - ☐ To make maps look nice
   - ☐ To ensure data storage is efficient
   - ☒ To detect and prevent errors before data use
   - ☐ To compress spatial files

7. **Which of the following would be a topological inconsistency?**
   - ☐ A point with the wrong coordinate system
   - ☒ A road segment not connected to the road network
   - ☐ A polygon with an incorrect name
   - ☐ A raster layer missing metadata

8. **Which ISO standard is mostly used to evaluate geographic information and spatial data quality?**
   - ☐ ISO 27001
   - ☐ ISO 9001
   - ☒ ISO 19157
   - ☐ ISO 14001

9. **A high RMSE (Root Mean Square Error) in spatial data indicates what?**
   - ☐ Strong alignment between actual and measured locations
   - ☐ High positional accuracy
   - ☒ Large deviation from true positions
   - ☐ Topological consistency

10. **In a spatial QA/QC process, what is the purpose of 'validation rules'?**
    - ☐ To correct visual symbology
    - ☐ To enforce attribute length
    - ☒ To ensure logical relationships between features are maintained
    - ☐ To update map projections

11. **What is your opinion regarding the statement: "*Data lineage refers to the spatial resolution of a dataset*"?**
    - ☐ True
    - ☒ False

12. **Which of the following techniques can help improve attribute accuracy during data collection?**
    - ☐ Manual sketching
    - ☒ Controlled vocabularies and drop-down menus
    - ☐ Free-text input only
    - ☐ Randomized data entry

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13. **What does *'logical consistency'* in spatial data refer to?**
    - ☐ The data matches a satellite image
    - ☐ The spatial and attribute data are in the same coordinate system
    - ☒ The adherence to rules of data structure and topology
    - ☐ The data is free from spelling errors

14. What is the main goal of a QA system in spatial data handling?

☐ To produce more data

☒ To identify and correct errors before data use

☐ To automate GIS functions

☐ To manage data storage

15. What is your opinion regarding the statement: "*Spatial resolution refers to the time frequency at which data is collected*"?

☐ True

☒ False

16. What is your opinion regarding the statement: "*A spatial dataset can have high positional accuracy but poor thematic accuracy*"?

☒ True

☐ False

17. Which of the following tools can be used to assess topological errors?

☐ Field survey equipment

☐ Raster calculator

☒ GIS topology checker (e.g., ArcGIS Topology Rules)

☐ Chart builder

18. Why is metadata important in spatial data quality?

☐ It makes the map look professional

☒ It documents the source, accuracy, and purpose of the data

☐ It increases file size

☐ It automates spatial analysis

19. Please, give an example of how you can control spatial data quality in the production phase (if possible, from your practice -you can select one of the quality elements):

During the data production phase, spatial data quality can be improved by applying simple validation steps within GIS. For instance, topology rules can be used to check that features are correctly connected and that there are no gaps or overlaps between polygons. In addition, using predefined attribute options, such as drop-down menus, helps reduce mistakes during data entry and ensures more consistent attribute values.

20. Please, give an example of how you can save costs with proper quality assurance (if possible, from your practice):

Proper quality assurance can help reduce costs by catching problems early in the data process. For example, checking the data during collection can prevent mistakes that would

otherwise need to be fixed later. If errors are only discovered after the project is finished, correcting them might require repeating parts of the work, which takes more time and increases costs. By applying QA steps early on, it is possible to avoid unnecessary rework and make the process more efficient.